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## **Becoming Certain About Deterministic Inversion Uncertainty**

Mike Bower, Hui Xie, Nestor Cuevas, Xiaobo Hong, Kent Harms, Joe Gremillion, and Mauro Viandante, SLB, Sugar Land, Texas, USA

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Logging-While-Drilling (LWD) tools have shown that electromagnetic (EM) measurements can be used to determine the subsurface's unknown electrical properties away from the trajectory of horizontally drilled wells. Modern LWD technologies can produce a model of the formation's electrical resistivity to depths of investigation (DOI) in excess of 150ft radially (Seydoux et al., 2014, Yankai, et al., 2018), in real-time.

In LWD applications, measurements of the different components of the magnetic field, excited by multiple loop antennas, are performed at various measured depths (MD) along the logging bottom hole assembly (BHA), at multiple frequencies of excitation. The magnetic-field components are then converted to deep and ultra-deep azimuthal resistivity measurements (DAR and UDAR respectively), which are used to estimate the resistivity distribution of the formation surrounding the wellbore via inversion. The resulting resistivity maps are interpreted to geological models and provide end-users with geosteering insights in real-time, or after drilling has been completed. The model is also used to estimate structural settings, productivity expectations, to evaluate fluid reserves or to plan additional wells.

Resistivity models are determined through different inversion methods (Tarantola, 1987, Meju, 2009), approximating the EM solution describing the acquired data. Accordingly, the development of more advanced inversion algorithms to determine resistivity models from LWD measurements while drilling, has received considerable attention (e.g., Li et al., 2005; Yankai et al., 2018; Thiel et al., 2018). These algorithms, which initially were developed to approximate a 1D local solution, first illustrated by Li et al., (2005) and have been increasing their dimensionality, to aid in the fitting of more complex data, recently moving to 2D solutions (Thiel, et al., 2016, 2018) and 3D (Omar et al., 2024).

### **Introduction**

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In geosteering applications, while determining the resistivity distribution is important, evaluating its accuracy is of equally relevance (Dong et al., 2015). Quantifying and analyzing the uncertainty of the inversion results undoubtedly helps to make confident geosteering decisions based on the interpretation of LWD data. Decisions may not be made if an acceptable level of uncertainty cannot be met.

To this end, stochastic algorithms have been developed to solve the inversion problems in 1D, thereby estimating an average of the model parameters and the uncertainty around them (Dong et al., 2015). However, stochastic methods require running multiple independent search instances, each randomly sampling thousands of formation models. Evidently, such random sampling of models can be increasingly taxing with higher complexity of the sought 1D structure, and therefore stochastic sampling is restricted to solving for a simplified model. This naturally leads to a lower vertical resolution output.

The alternative regularized parametric deterministic inversion algorithms yield higher resolution of 1D resistivity models (Yankai et al., 2018). Such regularized inversion algorithms will provide a series of accepted models inverted from a-priori random-sampled model structures, which yield a tolerable measure of data misfit. An average realization can be determined from the accepted models. To estimate the uncertainty of such regularized inversion methods, standard implementations call for the execution of multiple inversion runs, each departing from a random sample of the initial guess of the formation model. It is, however, debatable how many of such initial guesses are sufficient to properly sample the a-priori model space, and how far each initial guess can depart from the final outcome.

Thus, estimating uncertainty of the inverted formation profile continues to be a challenging problem. This is particularly so given the real-time requirements of LWD, versus the computational resources required to attain an ensemble of models which is representative of the statistical distribution describing possible solutions of the inversion problem. The onset of cloud computing has enabled such workflows to become more efficient, however they don't totally eradicate the problem and can be costly, especially with increasing complexity and dimensionality.

As an alternative to trying multiple initial guesses, other methods have been proposed to assess the uncertainty around the resulting inversion results, by exploring the posterior model space. Posterior sampling algorithms implement different random sampling of model space after the inversion, and further screening of the model realizations through a suitable accept/reject criterion, often based on ranking of the data misfit (Alumbaugh, 2000; Tompkins et al., 2011; Tompkins, 2012). At the outset, these methods yield a reduced number of forward simulations and thereby their computational cost is lower than running multiple inversions.

In this paper, we explore methods to estimate the uncertainty of a deterministic parametric inversion (DPI) of deep directional EM measurements (Habashy and Abubakar, 2004, Yankai et al., 2018), obtained within a 1D layered assumption of the formation. The first part of the paper will illustrate how the number of initial guesses affect the precision of the inverted results, and whether the widely used current number

of initial models is sufficient. Secondly, it will address the accuracy of the uncertainty, demonstrating how posterior sampling methodologies can help to reliably capture the uncertainty around these inverted results.

## Methodology

A synthetic model is used in the analysis, for which forward simulations of acquired data are performed as the well trajectory traverses the strata. At each MD, LWD azimuthal electromagnetic measurements are constructed from the simulated tool response with a realistic estimate of the tool noise. The tool noise is a complex model constructed from the bottom hole assembly (BHA) and the technical specifications of the LWD hardware. In turn, DPIs are performed to determine accepted models and an average model of the formation profile. The uncertainty of the formation model is then estimated using the models obtained by inverting the data starting from multiple initial guesses, and by posterior sampling the model space.

Figure 1 describes a 1D model used in our analysis, which consists of five parallel beds of different thickness, resistivity, and anisotropy. The wellbore is simulated at 85-degree inclination with the formation dip and azimuth being 0 degrees, leading to a relative dip of 5 degrees. This model is representative of a scenario where two resistive hydrocarbon-bearing reservoir layers are separated by a thin conductive and anisotropic shale layer. It also simulates a resistive hydrocarbon-filled body being located above a water-filled interval; thus the underlying medium is isotropic with vertical resistivity,  $R_v$ , and horizontal resistivity,  $R_h$  equal to 1 ohm.m. This is a common setting of hydrocarbon reservoir targets drilled using geosteering and reservoir mapping technologies.

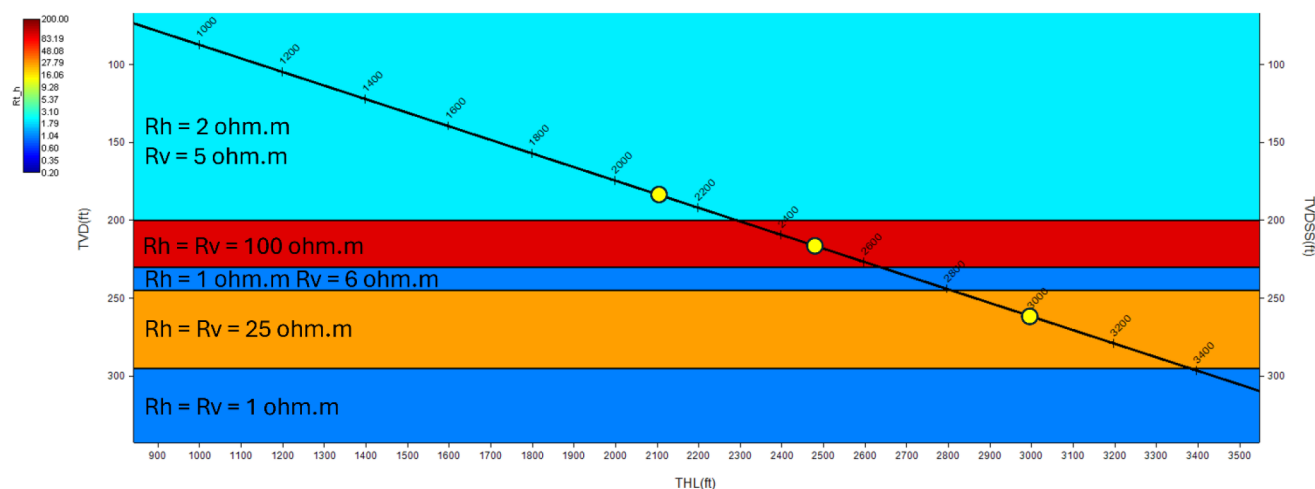


Figure 1—A 1D synthetic anisotropic model used in the study. Uncertainty of the inverted layer model is estimated at selected MDs, indicated by the yellow circles at 2100 ft, 2470 ft and 3000 ft.

Uncertainty of the inversion result is estimated first through analyzing the model ensemble resulting by starting from multiple initial guesses, hereby defined as "a-priori sampling" of the model space. Evidently, the strategy to define the model parameters of each initial guess is critical to minimize the cost function, whilst honoring the expected range of variability for the formation properties.

Common practices involve defining a reference model, for example derived from available well logs (Yankai et al., 2018) or alternatively randomly defined sampling within a-priori limits in layer resistivity, boundary position or numbers thereof, formation dip, etc. In this paper, we will focus on the randomly defined sampling approach in the following discussion. It is expected that increasing the number of initial guesses will increase the number of the accepted models. It is then debatable how the initial guesses should be sampled, while maintaining a reasonable computational cost.

The range of feasible inversion results can also be investigated by exploring the posterior model space, rather than launching multiple inversion runs with different initial guesses. Posterior sampling methods are well-known, and many strategies have been proposed (Alumbaugh, 2000; Meju, 2009; Tompkins et al., 2011; Tompkins, 2012) to expand the selection of model realizations distributed around the outcome of a DPI. The method developed by Tompkins will be referred to as "geometric sampling" going forward.

In building an ensemble of model realizations, some general criteria can be defined to yield at least plausible models. For example, one could expect that resistivity variations are constrained within reasonable limits, e.g.  $R_{min} \leq R_h$ ,  $R_v \leq R_{max}$ , or that  $1 \leq R_v/R_h \leq \lambda_{max}$ , etc. It is also anticipated that the resulting uncertainty should at least qualitatively agree with the expectations set by the physics of the LWD measurement. For example, sensitivity of the measurement to changes in the formation properties decreases with distance to the well (Omeragic D. et. al 2005). Thereby, it is expected that uncertainty increases for those layers located farther from the well. Conversely, the resistivity of those layers closest to the well should be more accurately defined. Also, it is expected that in an anisotropic medium, the estimated uncertainty  $\Delta R_v \geq \Delta R_h$ . Also, sensitivity to anisotropy within the EM measurements is highly dependent on the transmitter-to-receiver array spacing, the transmitted frequency, measurement used, and various elements of the resistivity environment (Bower et al, 2025).

An ensemble of models can be constructed to restrict the search space around the inversion result by building a geometrical description of the model space to satisfy the observations above. Such an approach was proposed by Tompkins et al. (2011) in the context of uncertainty estimation of 1D and 2D inversions in marine Controlled Source EM surveys. In this approach, the sampling of the model space is driven by the inverse of the linearized data covariance matrix, as evaluated at the last step of the inversion. From a qualitative standpoint, low covariance between model parameters yields low sensitivity of the data estimated when such model parameters are perturbed, and thereby, they are determined with higher uncertainty.

A second method is analyzed whereby the space of feasible posterior models is explored by drawing random model realizations, assuming a Gaussian distribution of the model space, centered around the inverted model criteria. This "direct sampling" method yields a collection of models that is further screened based on the same data-misfit criteria used in the inversion. This ensures consistency with the model selection strategy implemented on the ensemble, resulting in the a-priori sampling method. Screened models are then used to estimate the uncertainty of the inverted results. This method yields uncertainty of all model parameters, including resistivities, anisotropy, dip, etc.

In what follows, noisy LWD data simulated using the model described in Figure 1 are inverted to determine an average estimate of the formation's layered properties. Ensembles of models are then used to evaluate the  $p_{th}$  percentiles of the distribution function of each inversion parameter, for example:  $p_0$ ,  $p_5$ ,  $p_{15}$ ,  $p_{50}$ ,  $p_{85}$ ,  $p_{95}$ ,  $p_{100}$ , where  $p_0$  is the minimum resistivity model and the  $p_{100}$  is the maximum resistivity model.

## Uncertainty Analysis

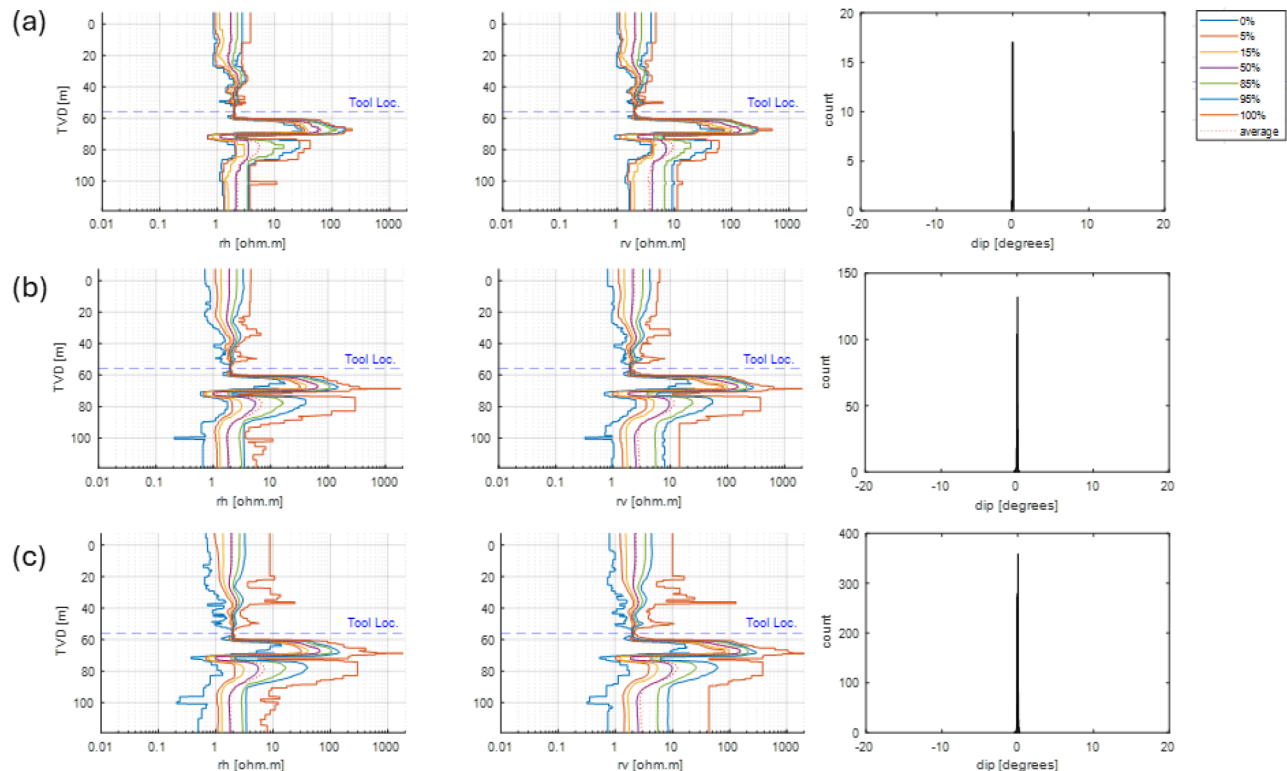
For the a-priori sampling, it is important to understand how many initial guess models are enough to satisfy the precision of the inverted results. Historically, one well-documented criticism of deterministic methodologies when compared to the stochastic workflow is that they may not sample enough model iterations, getting stuck in a local minimum and not accurately arriving at an accurate inverted result.

The first important discussion point is how those initial models are generated. If derived from a model guided approach (Yankai et al., 2018), there is a potential for the deterministic algorithm to converge in the wrong direction and therefore skew results based around unrealistic characteristics. However, if the models are sampled with enough randomness, and with wide enough constraints, then it could be deemed independent enough.

The second discussion point is how many of those randomly generated initial models are enough, especially in generic, or non-specialized geological cases. This number needs to be balanced along with

computational cost, mainly associated with time, since these inversion workflows need to return results in real-time drilling scenarios. This is in the order of seconds to a minute per inversion sample point. Previous suggestions have suggested that it might need to be in the region of 1000's of initial models, but this number may be as low as 50.

Figure 2 shows the uncertainty estimates obtained at MD 2100 ft (see Figure 1), running inversions with an increasing number of initial guesses,  $N=50$ , 500, 2000, and after removing outliers identified by misfit screening. Notice that the 50<sup>th</sup> percentile, the median of the ensemble, remains stable with the number of initial guesses. Those results have a slightly smoother estimate for a higher number of initial guesses. This indicates that  $N=50$  initial guesses should suffice to yield a representative average formation resistivity.



**Figure 2—Percentiles of the distribution of models obtained by increasing the number of initial guesses: (a) 50, (b) 500, and (c) 2000, for MD:2100ft.**

Despite having a larger number of initial guesses, the acceptance-to-rejection ratio remains constant. The overall spread across percentiles remains unchanged with the number of initial guesses, except the curves at the extrema ( $p_0$  and  $p_{100}$ ) exhibiting a spread with an increased number of initial guesses. This shows that the span of inverted models is bound by the a-priori distribution of initial guesses and the misfit acceptance criteria.

Additionally, the spread of the percentiles observed for the resistivity of the deepest conductive layer (i.e., TVD  $\sim$  100m) remains the same. In other words, the uncertainty doesn't change, despite the increasing distance to the well. This suggests that the sampling of deeper layers should span a broader range. A more suitable strategy requires further expanding the inversion range, however at the expense of increasing the computational costs of the inversion.

It may also be possible to increase the sensitivity to more distal features by adding additional transmitter-to-receiver arrays with longer spacing, or by altering the measurements used (measurement type or frequency) in the inversion process.

The results seen in Figure 2 therefore illustrate that with 50 initial guesses, a good result can be obtained with a reasonable precision, especially in a simple 1D environment. It is fair to state that in a more complex resistivity environment, more guesses may be required to accurately capture a representative model. However, as illustrated in Figure 3, increasing the number of initial guesses comes at a computation cost. Depending on how those model numbers are increased and on what format of machine the inversions are processed on, it can exponentially increase the time and therefore dollar cost of gaining little to no precision. It is therefore a balance between the precision and speed of which an answer is required.

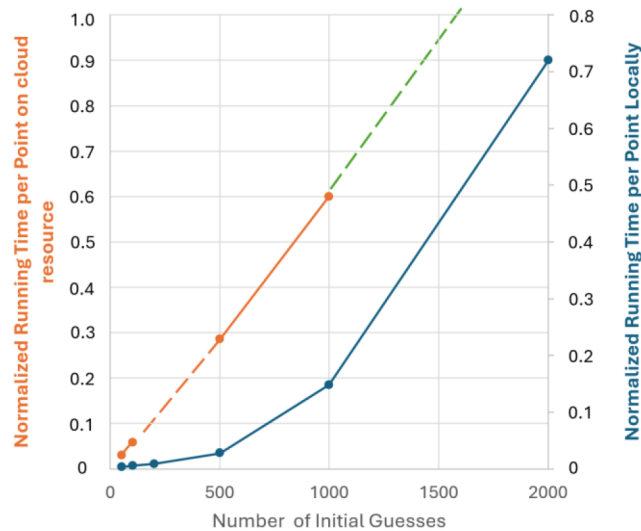
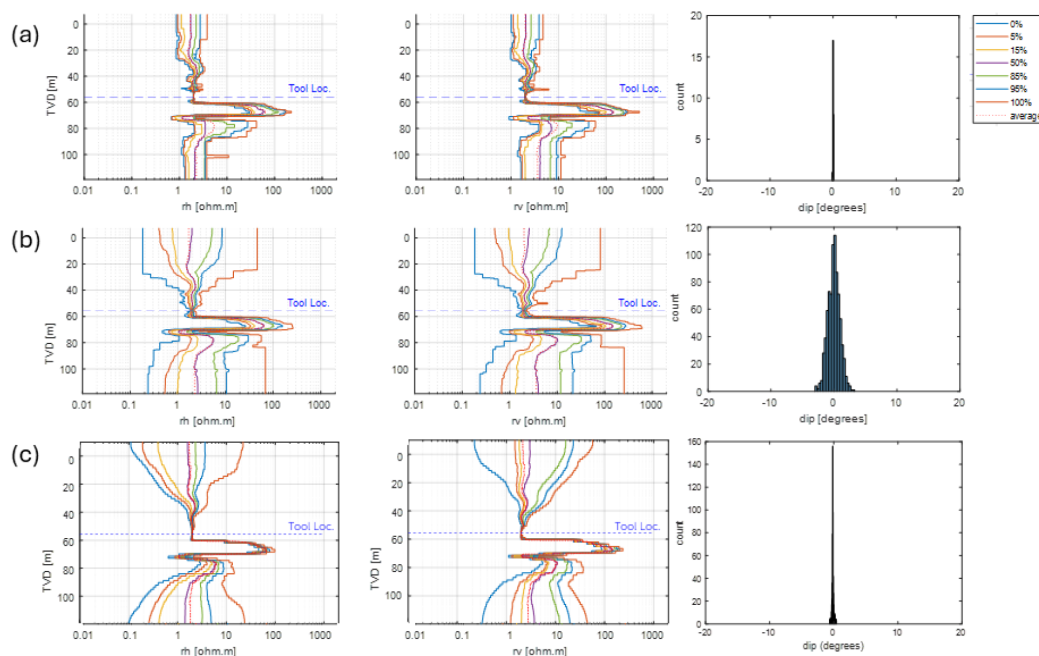


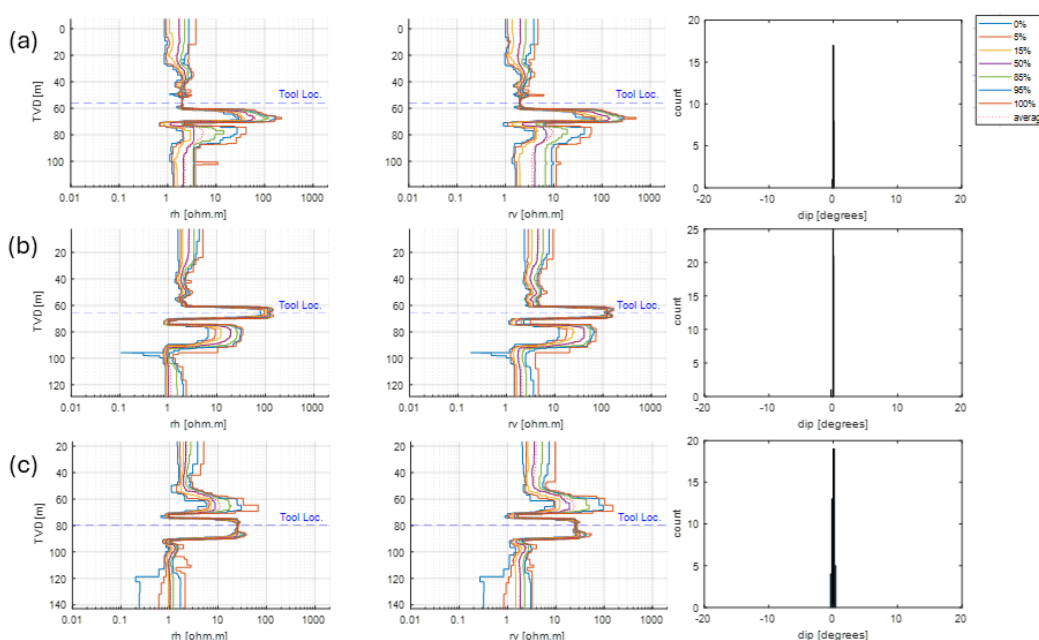
Figure 3—Illustration highlighting how computation speed varies between processing format and number of initial models.

In contrast, posterior sampling methods naturally yield a broader span of feasible models to be included in the ensemble. As an example, Figure 4 describes the percentiles of the distributions obtained at MD depth of 2100ft, using the direct (figure 4b) and geometric (figure 4c) sampling methods described above. They both exhibit a higher dispersion of the percentiles distally from the wellbore, as higher uncertainty is expected as a function of increasing distance from the well. Geometric sampling yields a narrow spread (lower uncertainty) of the percentiles estimated for the resistivity of the upper sand, driven by the higher sensitivity expected closer to the well. However, the resulting ensemble of models is centered on the inverted average formation and thereby the span of variability of such a resistive feature does not incorporate realizations of the isotropic, 100 ohm.m layer. In contrast, the ensemble of models obtained through a direct posterior sampling spans a wider range of realizations of the resistivity of the upper sand. Such a broader model ensemble yields a higher uncertainty of the layer properties.

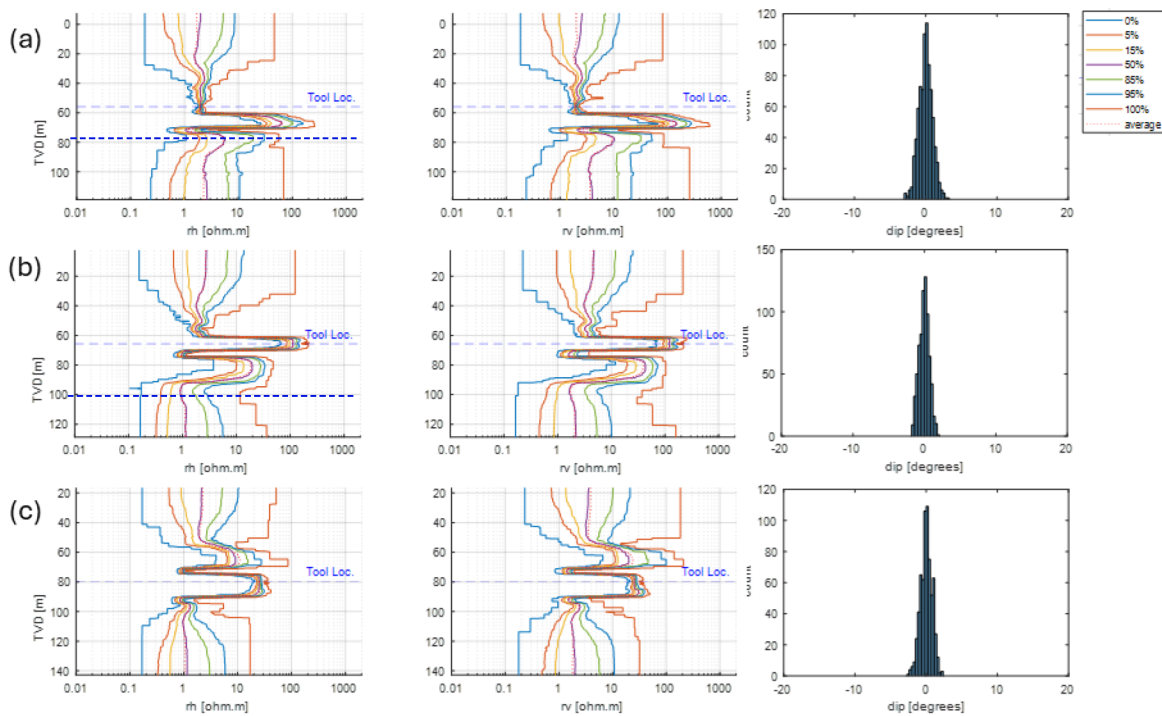


**Figure 4—Percentiles of the resistivity distributions obtained with different methods to sample the model space: (a) using 50 a-priori initial guesses, (b) by direct sampling, and (c) using geometric sampling at 2100 ft MD (see figure 1)**

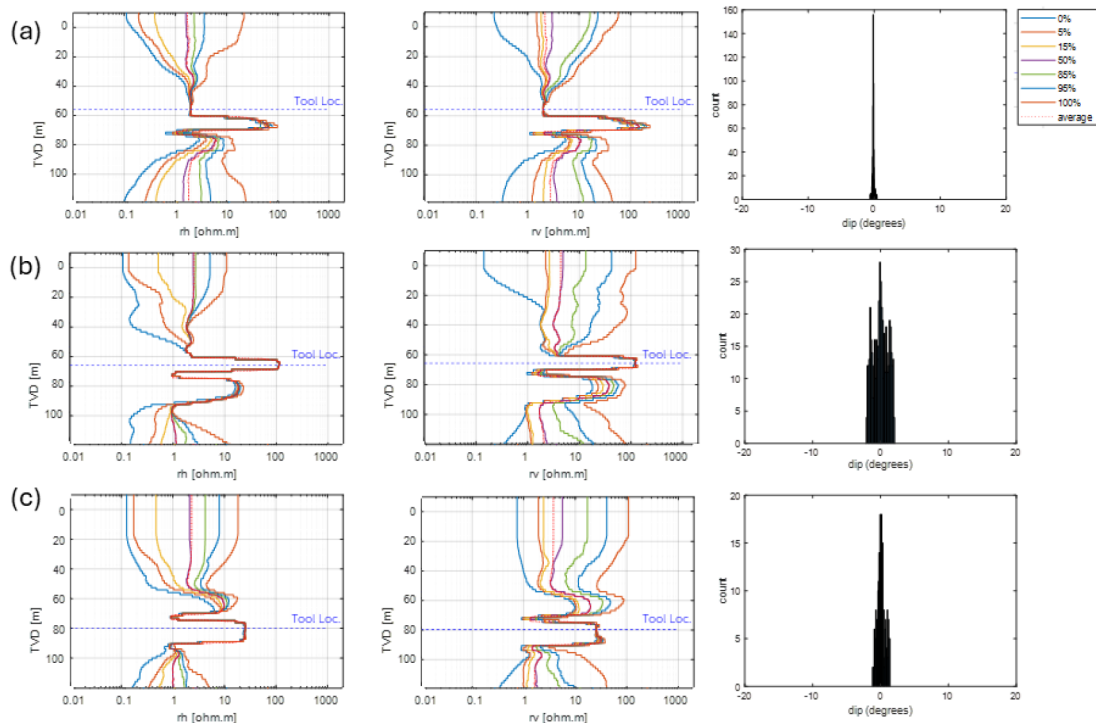
As a function of measured depth (MD = 2100, 2470, 3000 ft), the estimate of uncertainty of the inversions is shown in Figure 5, Figure 6, and Figure 7, using the a-priori initial guess, the direct posterior sampling, and the geometric posterior samplings methods respectively. It is expected that the lowest uncertainty should be observed where the tool is. This is best observed on the ensembles obtained using the a-priori initial (N=50) guesses (Figure 5) and that obtained using the posterior geometric sampling (Figure 7). The direct posterior sampling (Figure 6) yields a wider spread of the percentiles at the layers where the well trajectory is located.



**Figure 5—Uncertainty estimates for an ensemble determined by using a 50 a-priori initial models, as a function of MD: (a) 2100 ft, (b) 2470 ft and (c) 3000 ft.**



**Figure 6—Uncertainty estimates for an ensemble determined with a direct sampling method, as a function of MD: (a) 2100 ft, (b) 2470 ft and (c) 3000 ft.**



**Figure 7—Uncertainty estimates for an ensemble determined with a geometric sampling method, as a function of MD: (a) 2100 ft, (b) 2470 ft and (c) 3000 ft.**

An added benefit of posterior sampling methods is the capability to "interrogate" the model ensemble to determine the uncertainty of each model parameter, independently of the inversion process. Figures 6 & 7 show how the posterior sampling can also be used to sample feasible realizations of formation dip, as is evidenced by the wider uncertainty in the dip histogram for each figure's subplot. In this case, all

model parameters are sampled simultaneously, however, one could also sample feasible realizations of an independent parameter, given suitable assumptions for the other parameters.

It is worth noting that all three methodologies discussed would derive different uncertainties or probability distributions if the inputs to the inversions were modified (e.g. curve type, frequency, transmitter-to-receiver array spacing etc.). It is important because it helps to understand if the uncertainty found in the results is an outcome of the true uncertainty in the measurements and inversion sampling method or is a result of the inversion being unable to fit low quality input data.

In this analysis, one-inversion has been applied to the results throughout the model. It is important to understand that in a real-life scenario, the input required for landing the well as in the 2100ft MD example, could differ from those required in the inversions at 2470 ft MD and 3000 ft MD due to the differing resistivity context of the sample MDs. Consequently, this could lead to a different range of uncertainty results.

Furthermore, these results have all been carried out on a simple 1D model. However, should the 1D assumption of the measurements break down, the implementation of a 1D inversion will lead to artifacts within the models and potentially biasing the estimated uncertainty, no matter which methodology is applied. Therefore, it is imperative that the appropriate dimensionality of inversion be applied to fit the resistivity context, to maximize the results with the lowest uncertainty possible.

Ultimately these results show that the methodology employed to understand the uncertainty in the inverted measurements can yield varying results, depending on the theoretical assumption and the definition of the sampling space. Such variance in uncertainty results may influence decisions made in a real-time geosteering context, and this is why it is important to understand if a thorough methodology has been applied which sufficiently captures all the uncertainty present under reasonable assumptions.

The data presented here proves that it is possible to vary the amount of uncertainty, however, it is also important to recognize that there is an irreducible uncertainty in the measurements used as the input to these inversions. Understanding the uncertainty of the measurements as shown in [Xie et al., \(2024\)](#) can help to understand how that may also amplify the uncertainty in the inversion results, further impacting real-time decision making. To reduce the risks associated with inputting noisy measurements, the best possible data is required.

In the forward modelling stage, a robust noise model should be utilized to accurately capture and quantify the measurement uncertainty present and manage a user's expectations. When utilizing real data, a tool architecture with as small noise level as possible is advised.

## Conclusions

In geosteering applications quantifying the accuracy of the deterministic parametric inversions can be a challenging problem. Regularized optimization algorithms converge into a single realization of the space of feasible solutions to the inversion, that describes on average the spatial resistivity distribution of the formation around the well trajectory. The standard practice calls for running multiple inversion processes starting from diverse initial models. The ubiquitous question seeks a rule of thumb for the required number of initial models. Herein we show that increasing such number makes little impact to the estimates of uncertainty if the limiting bounds of the span of a-priori is not itself randomized.

We further demonstrate that enhancing the ensemble of feasible 1D models can be attained by posterior sampling of the model space, avoiding the execution of lengthy inversion processes, but rather through forward simulation and misfit testing of other models that can satisfy the LWD data. In this analysis we have implemented two methods of posterior sampling, based on different strategies to define a distribution of models centered on average formation resulting from the DPI inversion. We demonstrated that these posterior sampling methods better reflect the expected behavior of uncertainty as a function of distance to the well, and increased uncertainty of the vertical resistivity component in an anisotropic medium.

Notice that the main benefit of posterior sampling methods is the reduction of the computational cost, essentially dependent on the number of forward evaluations. Therefore, it is anticipated that these methods can be applied in a real time LWD setting, to estimate uncertainty of the inversion in 1D, 2D and 3D problems.

Furthermore, this posterior sampling concept can be used to query the distribution of feasible models to investigate uncertainty to specific variables describing the model. For example, for an assumed resistivity vertical profile (e.g. given by the median, or weighed average), posterior sampling can be used to study independently the expected dispersion of formation dip, layer thickness, etc.

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